

Preferential flow patterns in forested hillslopes of east Tibetan Plateau revealed by dye tracing and soil moisture network

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Abstract

Preferential flow (PF) through soil and regolith results in a rapid vertical and lateral water movement within the profile. This study focused on quantifying PF in contrasting forested hillslopes of the Hailuo Valley, located on China's east Tibetan Plateau. Quantifying PF in this region is challenging since the underlying matrix is complex, with shallow soils, thick underlying saprolite, and the combined effect of moraines and weathered residual material. We developed new methods that integrated dye tracing with a high-frequency soil moisture monitoring network to characterise PF pathways and quantify PF frequency on contrasting forested hillslopes (coniferous and broadleaved). Dye tracing experiments showed that soil-root interfaces in the upper soil layer (10–50 cm) and soil-rock interfaces in the deeper soil layer (50–100 cm) were the primary PF pathways and that large rocks strongly influence the percolation depth of water. The high-frequency soil moisture network revealed that the mean PF frequency was 67.8% and 71.7% in the coniferous and broadleaved forests, respectively. The frequency of PF in the deeper soil layer increased from upslope to the downslope locations on both forested hillslopes, highlighting the tendency for PF to occur downslope. In addition to matrix conditions (e.g., stony saprolite soil), the total amount and maximum intensity of precipitation (as throughfall in this study) were identified as factors that control PF occurrence. In contrast, the initial soil water conditions (at 10, 50 and 100 cm depth) were insignificant in predicting PF occurrence. Results of this field-based study highlight that PF is a ubiquitous and critical subsurface flow mechanism that regulates rainfall-runoff relationships in forests of the Tibetan Plateau, underscoring the need to consider PF in hillslope hydrological modelling.

Highlights

- Combined dye tracing and soil moisture monitoring network were applied to explore PF on forested hillslopes.
- Root networks and rock fragments constituted the main PF paths.

- PF frequency increased from upslope to downslope in two forest types.
- Initial soil moisture conditions showed no controlling effect on PF occurrence.

KEYWORDS

critical zone, flow pathways, hillslope hydrology, hydropedology, subsurface flow, subsurface structure, throughfall

1 | INTRODUCTION

Preferential flow (PF) in soils refers to the process by which infiltrating water bypasses most of the soil matrix, resulting in a rapid vertical and lateral movement of water within the soil profile, typically delivering water to deeper horizons (Beven & Germann, 1982; Guo & Lin, 2018; Jarvis et al., 2016; Nimmo, 2021). Hydrologically, PF affects the water balance of the critical zone - influencing the rate and pathways of infiltration, runoff, subsurface stormflow, and groundwater recharge (Beven & Germann, 2013; Buttle & McDonald, 2002; Dusek & Vogel, 2016; Green & Erskine, 2011; Guo et al., 2020; Jost et al., 2012). Ecologically, PF influences the mass transport of water and associated loadings of solutes, sediments and nutrients (Clothier et al., 2008; Fishkis et al., 2020; Larsbo et al., 2014; Liu & She, 2020; Weill et al., 2011; Wilson et al., 2008). Given the strong influence of macropores on the flow paths and residence times of water in the soils, PF could play an essential role in the dynamic feedback between water and vegetation (Guswa et al., 2020). Understanding PF and its variability over space and time will advance fundamental knowledge of coupled hydrological and ecological processes (McDonnell & Beven, 2014; Nimmo, 2021).

Forest soils commonly have high infiltration capabilities that favour PF, given extensive macropores systems consisting of decayed root channels and animal burrows in shallow soils (Alaoui et al., 2011; Beven & Germann, 1982; Demand et al., 2019; Laine-Kaulio et al., 2015; Wiekenkamp et al., 2016). Forested hillslopes often have a variably weathered and deposited zone between the soil surface and the intact bedrock (Salve et al., 2012). Rainwater can infiltrate into the weathered and deposited zone through preferential pathways before reaching the bedrock, and the presence of macropore networks is known to influence the transport of water (Guo et al., 2019; Salve et al., 2012). Conceptual understanding of PF in the weathered and deposited zone continues to evolve. In steep hillslopes with shallow soils and thick saprolite, PF-induced subsurface flow through macropore systems is a potential link to rapid runoff response (Sidle et al., 2001). Shallow soil promotes lateral development of root networks and

channels, delivering the water downward at a high velocity. Thick saprolite may promote rapid water transport through structural macropores or cracks, strongly affecting the water residence time in soils (McGlynn et al., 2002). Saprolite underlying hillslope areas is typically stonier and coarser-textured than toeslope colluvium (Hartmann et al., 2020; Laine-Kaulio et al., 2015). Coarse-textured soils typically have no prominent structural channels and have high saturated hydraulic conductivity along the soil profile, favouring PF occurrence (Hardie et al., 2011; Hendrickx & Flury, 2001; Laine-Kaulio et al., 2015). Stony soils alter water flow by dispersion, lead to an unstable wetting front due to the impermeability of rock, and are markedly associated with the PF process (Bogner et al., 2014; Cerdà, 2001; Jarvis, 2007; Liu & She, 2020). The ubiquity of PF in forests is of great importance to hillslope hydrology, as PF pathways can be a significant factor in predicting subsurface flow characteristics on hillslopes (Wienhöfer & Zehe, 2014). Thus, PF and its controlling factors should be carefully examined (Beven & Germann, 2013; Li et al., 2018).

Numerous methods have been developed to visualise and quantify PF at different spatial and temporal scales (Allaire et al., 2009; Jarvis et al., 2016). There have been great advances in X-ray and magnetic resonance imaging (MRI) methods for imaging macropore structures at soil core or soil column scales (Feng et al., 2020; Jarvis et al., 2017). Dye tracing experiments were frequently used to reveal PF patterns in soil profiles (Filipović et al., 2020; Laine-Kaulio et al., 2015; Weiler & Flühler, 2004). Moreover, the increasing availability of geophysical techniques (e.g., ground-penetrating radar and electrical resistivity tomography) offered new opportunities to quantify PF with minimal disturbance at plot to hillslope scales (Gormally et al., 2011; Guo et al., 2014; Guo et al., 2020; Haarder et al., 2011; Koestel et al., 2009). At catchment and regional scale, reliable measurements and quantification of PF by the methods mentioned above remain arduous, with a need for bridging gaps across scales (Guo & Lin, 2018; Jarvis et al., 2016; Nimmo, 2012; Nimmo, 2021). An alternative method utilizing a high-frequency soil moisture monitoring network was developed, which has advanced in-situ detection of PF across spatial and temporal scales (Graham & Lin, 2011; Wiekenkamp

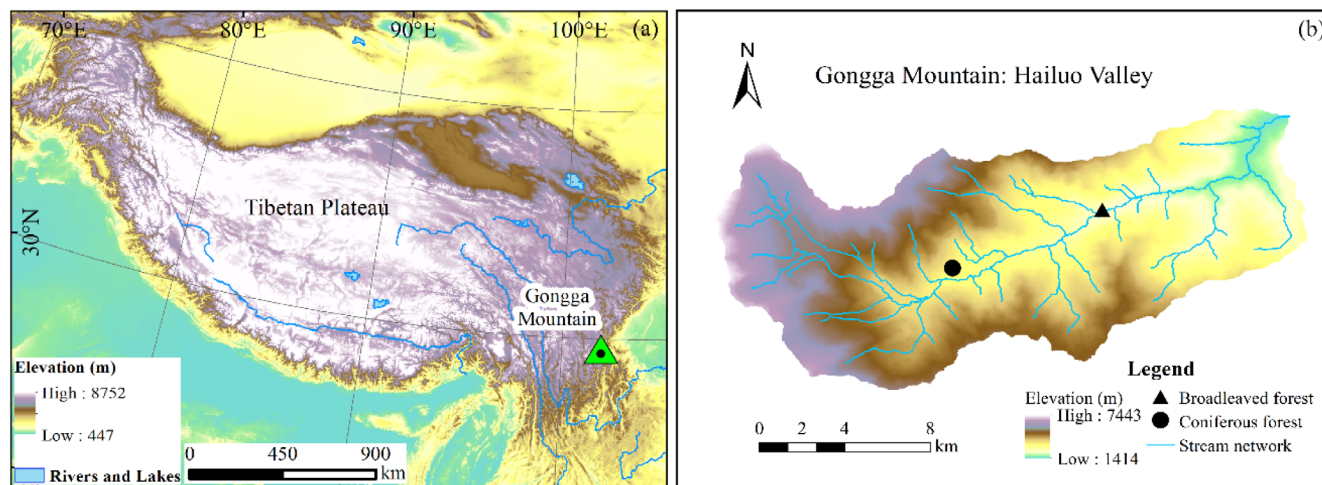


FIGURE 1 Locations of (a) the Gongga Mountain and (b) the studied forested hillslopes

et al., 2016). Using this method, researchers have gained a new understanding of PF relationships with watershed characteristics, such as rainfall events (Jost et al., 2012; Lozano-Parra et al., 2015; Tian et al., 2019), soil depth and hillslope topography (Graham & Lin, 2011; Liu & Lin, 2015; Wiekenkamp et al., 2016), and initial soil water conditions and landforms (Demand et al., 2019; Holden, 2009; Liu & Lin, 2015; Tang et al., 2020). However, using a data-intensive network-based approach alone cannot resolve individual PF pathways or establish the continuity and possible soil depths of the PF processes. Here, we propose that integrating dye tracing experiments with a high-frequency soil moisture monitoring network is a promising way forward to quantify PF characteristics and its contribution to hillslope water cycling in forests. We use the combined approaches to illuminate multiscale PF processes and their controlling factors under the natural conditions at field sites.

To advance the understanding of PF in steep mountain forests, this study focused on PF patterns on forested hillslopes in the Hailuo Valley, located in the east Tibetan Plateau, China. Quantifying PF and its contribution to hillslope water cycling in this region is challenging since the underlying porous media matrix is complex. The forested hillslopes have shallow soils, thick underlying saprolite, and the combined effect of moraines and weathered residual material. Our specific objectives were to (i) quantify PF occurrence over space and time, using methods that integrate intensive measurements from soil moisture monitoring networks with field-scale dye tracing experiments, (ii) identify PF pathways and frequency in soil profiles of contrasting forest woodland habitats (coniferous and broadleaved), and (iii) explore factors that potentially regulate PF patterns. Our results provide new insights into PF in forests on the Tibetan Plateau.

2 | MATERIALS AND METHODS

2.1 | Site description

Our field sites are located in the Hailuo Valley on the eastern slope of Gongga Mountain of the east edge of the Tibetan Plateau (Figure 1). Elevation along the east slope of Gongga mountain rises by 6400 m along just 29 km of horizontal distance, resulting in distinct vertical zones of climate and a rich natural vegetation diversity along the elevation gradient. The climate is dominated by the southeast Pacific monsoon, with annual precipitation concentrated during the wet period from May to October (Zhong, 2002). Two hillslopes were selected for explorations of PF, in contrasting forest types along the vertical vegetation gradient of the Hailuo Valley. One is in the evergreen coniferous forest (*Abies fabri*, *Picea brachytyla*) at 3000 m asl, and the other is the deciduous broadleaved forest (*Lithocarpus cleistocarpus*, *Salix babylonica*) at 2260 m asl. On these hillslopes, soils are characterised by thin O and A horizons, a thick B horizon, and rich rock fragments (Figure 2) with the occasional size of a single rock fragment up to meters in diameter (He et al., 2004).

2.2 | Soil measurements

2.2.1 | Soil characteristic measurements

To quantify soil properties of hillslopes, soil samples at three slope positions (upslope, middle and downslope) and depths (10, 50 and 100 cm) were collected in summer 2020, at locations within 2 m of the installation positions of the soil moisture sensor arrays. Sampling included

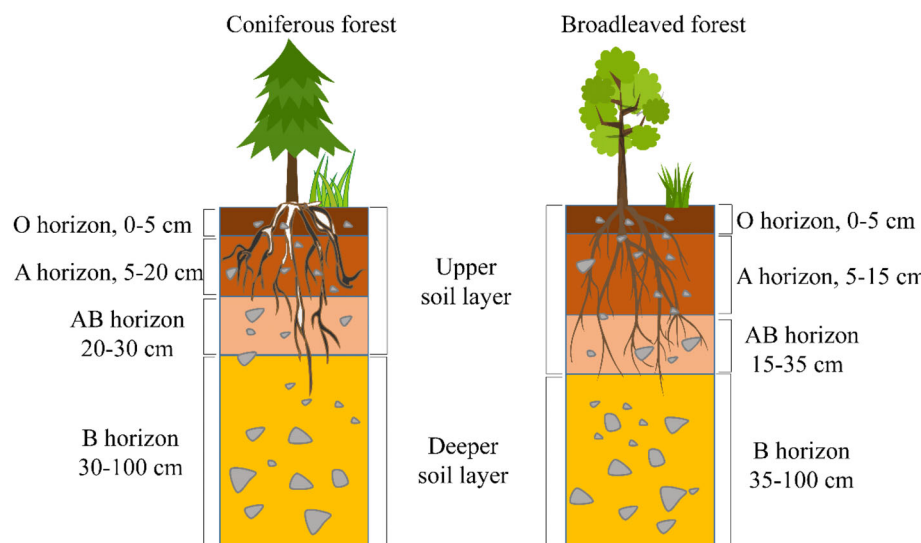


FIGURE 2 Conceptual soil profiles in the coniferous and broadleaved forests (not to bedrock)

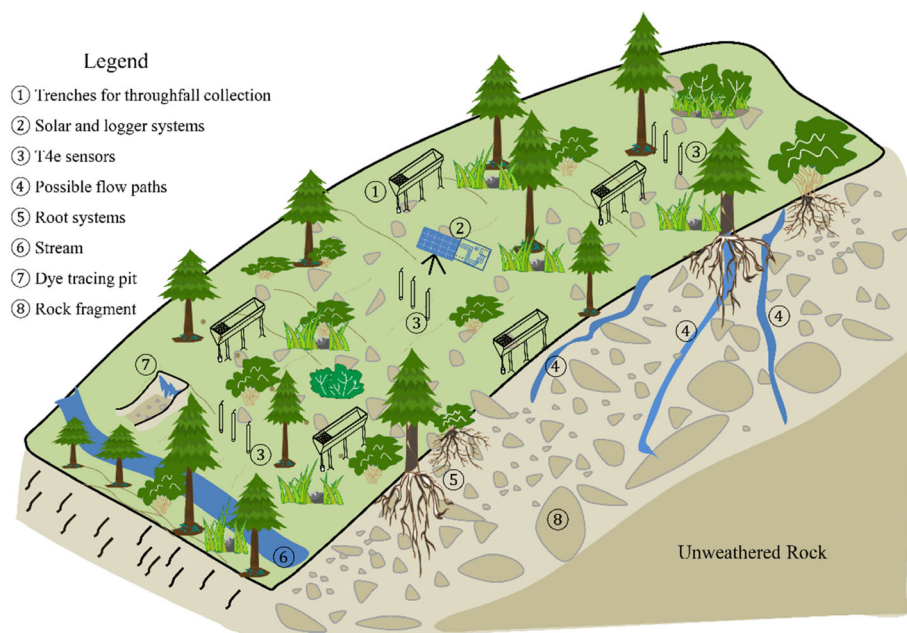


FIGURE 3 Schematic diagram of the studied hillslope and the hydrological monitoring equipment (not to scale)

undisturbed soil core samples ($n = 3$ for each depth, cutting ring 100 cm^3 in volume) and disturbed soil ($n = 3$ for each depth). Root biomass was collected from a $50 \text{ cm} \times 50 \text{ cm}$ quadrat ($n = 2$ for each slope position) at the upper (10–50 cm) and deeper layers (50–100 cm). Soil core samples were oven-dried at 105°C for 24 h to determine soil bulk density, and soil porosity was calculated from the dry bulk density with a particle density of 2.65 g/cm^3 (Shao et al., 2006). Oven-dried soil core samples were later passed through a 2-mm sieve to separate rock fragment ($>2 \text{ mm}$), and rock fragment content (cm^3/cm^3) was determined by the volume replacement method. Disturbed soil was divided into two sub-samples after being air-dried and passed through a 2-mm sieve, with one for soil particle size composition analysis by

laser particle size analyser (Mastersizer 3000, Malvern Instruments Ltd, UK) and the other for soil organic carbon (SOC) by dry combustion method after passing through 0.01 mm sieve (Vario Macro Cube, Elementar Analysensysteme, Germany). Soil organic matter (SOM) was estimated as SOC multiplied by 1.72 (Pribyl, 2010). Collected roots were gently washed to remove attached soils and oven-dried at 60°C for 72 h to constant weight. Soil saturated hydraulic conductivity (K_s) was measured using a Guelph Permeameter (Soil Moisture Equipment Corp, USA) with at least three depths (3 replications at each depth) in the upper (10–50 cm) and deeper soil layer (50–100 cm). Detailed processes of measurement refer to the Guelph manual (Guelph Manual, Soil Moisture Equipment Corp, USA).

2.2.2 | Soil moisture monitoring

A soil moisture monitoring network was deployed at three slope positions (upslope, middle and downslopes) and depths (10, 50 and 100 cm) on the forested hillslopes (Figure 3). Soil water potential (T4e, UMS Corp, Germany; measurement ranges from -85 kPa to 100 kPa with 0.03 kPa precision) was used to replace generally used soil water content, and soil water potential is the force that governs the water transport in soils (Novick et al., 2022). Installation of sensors was processed as follows: a nearly 60 cm-wide and 120 cm-deep soil profile was firstly excavated, and then holes were drilled at 10 , 50 and 100 cm depth with an inclination between 15° – 20° in the soil profile. Next, a total of 18 sensors (3 depths $\times$ 3 slope positions $\times$ 2 forest types) were carefully installed. Finally, soil was backfilled with the original soils layer by layer to maintain the original soil setting and minimum disturbance. The installation was finished at the end of October 2019.

2.3 | Dye tracing experiments and image analysis

2.3.1 | Dye tracing experiments

Dye tracing experiments were conducted at the downslope during summer 2020 using a double ring method

(Figure 4). A dye tracer solution (Brilliant Blue FCF, 4 g/L) was continually added to reach a constant head of 2 cm, and a total of 16 L solution was added for each experiment (Figure 4a). After 4 h, the tracer completely infiltrated into the soil, and the soil profile was ready for excavation. The dye-stained area was separated into two sub-areas along the diameter of the outer ring (Figure 4b). Half of the dye area was excavated from the soil surface to deeper soils for horizontal dye photographs until there was no clear dye blue colour in the soils (Figure 4c). The other half of the dye area was excavated from 45 cm away from the diameter center towards the double ring center for vertical dye photographs (Figure 4d). The excavation interval of soil profiles was set to 5 cm. Dye photographs of soil profiles were photographed by a D700 camera (Nikon Corp, Japan).

2.3.2 | Preferential flow imaging

Image processing was conducted based on the theory established by Flury et al. (1994). First, a geometric correction was applied to eliminate geometric distortion caused by photographing. Second, colour adjustment was applied to simplify image information and highlight the stained area. Third, RGB images were converted to gray-scale images, and a cross-validated threshold of the colour scale was set to turn images into binary images with the stained area as black and the unstained area as white.

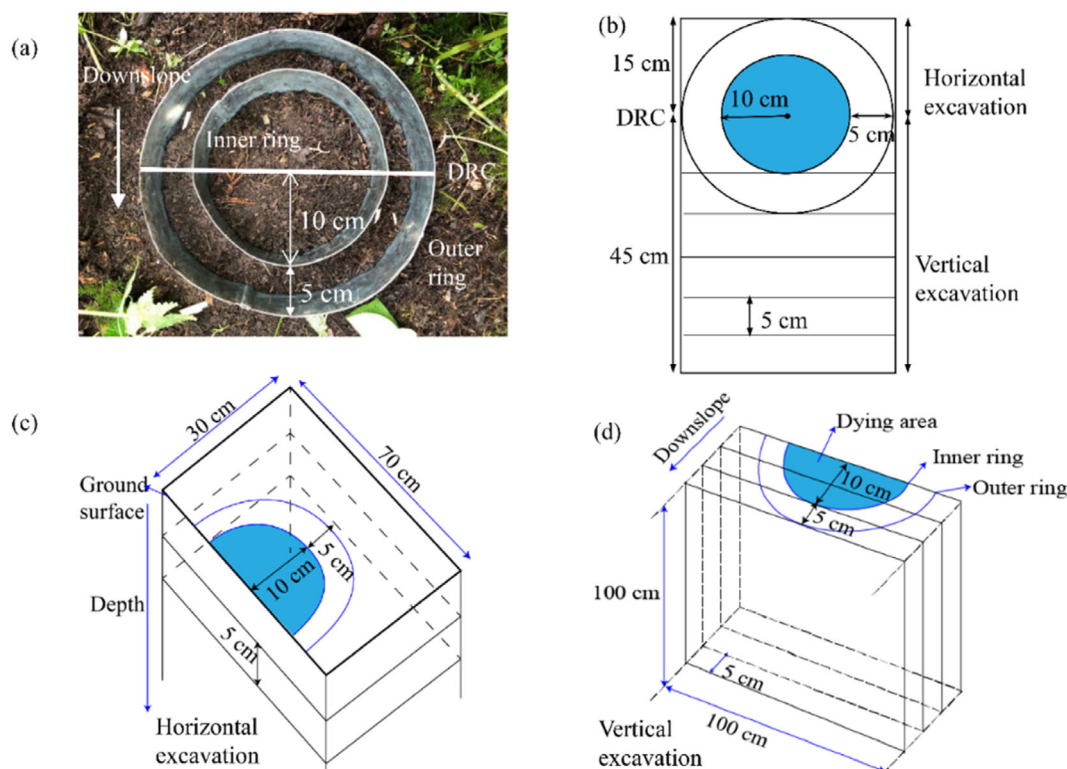


FIGURE 4 Double ring method for dye tracing experiment (a) and schematic diagram for dye area and field excavation (b) in horizontal (c) and vertical directions (d). (DRC denotes double-ring center, not drawn to scale).

The following parameters derived from the dye tracing experiments were chosen for detection and quantification of PF patterns (van Schaik, 2009): (i) dye coverage (stained area divided by 30×70 cm for horizontal photographs, 100×100 cm for vertical photographs), the best guess of total infiltration into the profile, where a higher infiltration capacity will result in a larger stained area; and (ii) the maximum depth of blue stains (MDS), the maximum penetration depth of water in the PF pathways. Here we took the total depth of the last stained soil layer visualised in the vertical soil profile as MDS.

2.4 | Preferential flow occurrence

2.4.1 | Throughfall measurements

Rainfall on the forested hillslopes was designated as throughfall, i.e., the effective throughfall reaching the land surface after canopy interception. Five throughfall collection trenches ($300 \text{ cm} \times 25 \text{ cm} \times 30 \text{ cm}$) were placed 1 m above the ground with a slight inclination (less than 2°) to eliminate the non-uniform distribution of throughfall (Li et al., 2010). The outlet of the trench was connected to tipping buckets by pipes for throughfall amount measurements. A throughfall event was defined with a minimum amount of 1 mm. The end was defined as the last monitored response of a tipping bucket followed by 6 h with a throughfall amount of less than 0.2 mm. T4e sensors and tipping buckets were all connected to a data logger (CR100, Campbell Scientific Ltd, USA) for data collection at the 15-min time interval to capture the dynamics of soil moisture. Recorded data during the growing season (May–September) in 2020 was used in further analysis. The start time of the soil moisture monitoring network and throughfall was 6 months after the installation of sensors, excluding possible disturbance and freezing damage during the cold season (November–April).

2.4.2 | Preferential flow quantification

The throughfall and soil moisture data at the 15-min time interval were used in PF analysis. Based on the response time series of sensors through the soil profile to a throughfall event, PF has been divided into two ways: (1) a non-sequential response to a throughfall event, where a deeper soil layer responds before shallow soil layers (Lin & Zhou, 2008), and this indicates that rainwater bypassed the shallow soil layers and percolated into the deeper soils; and (2) a rapid rainwater flux through the monitoring soil profile, exhibiting a sequential response of sensors with depth at a velocity greater than the measured K_s of

adjacent soil layer (Beven & Germann, 1982; Flüßler et al., 1996). Once the throughfall events were determined, if the soil water potential increased above the threshold (0.3 hPa), the response times of sensors were recorded and compared. Sequential and non-sequential responses were determined as follows (Figure 5). Non-sequential responses (NSR) were defined when sensors responded out of sequence, i.e., any deeper sensor responded earlier than the upper sensors (e.g., the 50 cm sensor showed a response before the 10 cm sensor). Sequential responses (SR) were defined when sensors responded in sequence, with the surface sensor responding first, then the next deeper sensor, and so on through all sensors in the soil profile. According to the PF definition, all NSR events and part of SR events were PF events. Further, the sum of PF events and SR events is referred to as total response events. The PF frequency was calculated by the following equation:

$$\text{PF frequency} = \frac{\text{PF events}}{\text{PF events} + \text{SR events}} * 100\%$$

2.4.3 | Factors controlling preferential flow

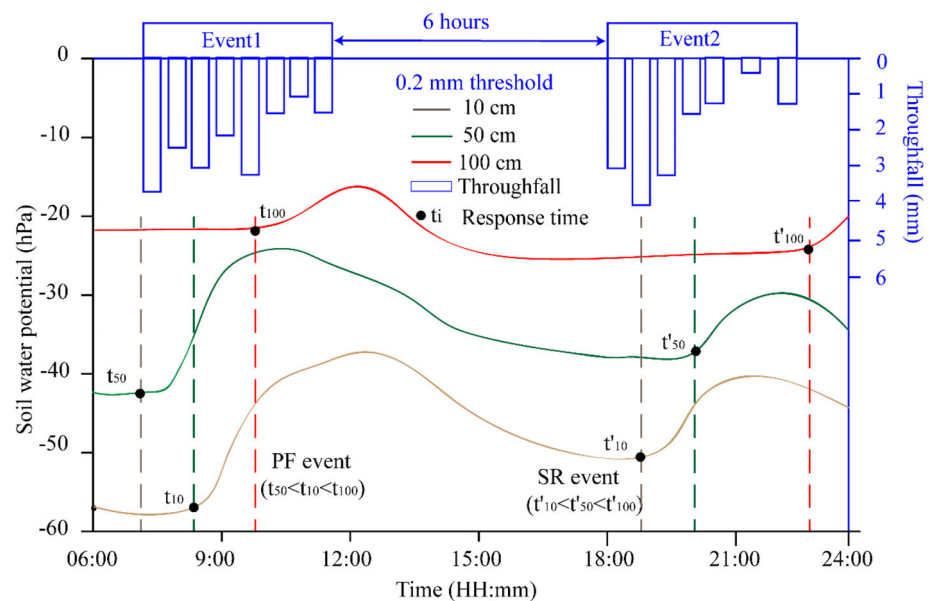
To explore potential factors that control PF occurrence, temporally varying factors including throughfall characteristics (total throughfall amount, throughfall duration, throughfall intensity, maximum throughfall intensity, throughfall skewness, kurtosis, and standard deviation) and initial soil water conditions, and spatially varying factors including soil bulk density, soil porosity, SOM, clay content, rock fragment content, and root biomass were explored as potential controlling factors (Guo & Lin, 2018; Liu & Lin, 2015; Wiekenkamp et al., 2016). Statistical t-tests were used to compare factors and to generate hypotheses about their potential significance in generating PF. Factors for PF-producing events vs. SR-producing events were tested to indicate whether there were any significant differences ($p < 0.05$). When a t-test is significant, the factor analysed is considered to influence PF processes (Liu & Lin, 2015). We further merged values of different slope positions into a group to explore differences at the hillslope scale.

3 | RESULTS

3.1 | Soil properties of studied hillslopes

Many of the soil properties between the upper and deeper layers were significantly different ($p < 0.01$) in both forests (Table 1). Results showed that the upper soil layer

FIGURE 5 A schematic graph of individual throughfall event delineation and PF detection



had higher SOM (mean value of 52.81 and 52.19 g/kg in the coniferous forest and broadleaved forest, respectively), lower bulk density (mean value of 0.89 and 0.92 g/cm³ in the coniferous forest and broadleaved forest, respectively) and more root biomass (110.33 and 215.33 g in the coniferous forest and broadleaved forest, respectively) in both forests. Moreover, the soil at the downslope location had the highest SOM content in both forests. The upper soil layer has higher K_s compared with that of the deeper soil layer and showed considerable changes in K_s even at nearby sites.

3.2 | Preferential flow pathways

Dye photographs clearly showed the important effect of roots and rock fragments on PF pathways (Figure 6). For the horizontal dye patterns in the coniferous forest (Figure 7), a sharp decrease in the dye coverage was observed at 20 cm depth, where the existence of rock fragments drove water to flow through the soil-rock interfaces and bypassed the soil matrix (Figure 6c). Meanwhile, the surging increment of the dye coverage at 85 cm depth was likely due to small-sized rock fragments, where the dye solution flowed along lacunar pores and stained the surrounding soil matrix and rocks, leading to a larger stained area. In the broadleaved forest, a sharp increase of dye coverage was observed at 20 cm, where root networks drove water flow laterally and stained surrounding soils along the flow pathways (Figure 7), and then the dye coverage fluctuated at the depth between 20 and 50 cm, where the complex soil-root-rock structures were observed during soil profiles excavation. Below 50 cm depth, dye coverage decreased with the increase of soil depth.

For the vertical dye photographs, an increasing trend of dye coverage from far to near the dye ring center was observed in both forests (Figure 8). In addition, soils in the broadleaved forest showed a higher mean dye coverage, indicating a more extensive lateral movement. Results also demonstrated that the broadleaved forest had faster water velocity and further flow distance compared with that of the coniferous forest (Figure 8) because the topography of dye tracing experiment sites in the two forests was similar.

The maximum depth of stain (MDS) was 115 cm in the coniferous forest and 95 cm in the broadleaved forest. Soil profile excavation showed that the larger MDS in the coniferous forest was mainly controlled and regulated by the large rocks in the deep soil layer where water with tracer flowed fast through the soil-rock interfaces to deeper soil layer.

3.3 | Throughfall events and soil moisture conditions

Throughfall was characterised by long duration and low intensity in both forests. A total number of 70 and 45 throughfall events were delineated with an average duration of 27 and 21 hours, average throughfall intensity of 0.53 and 0.55 mm/h, and mean total event precipitation amount of 17.14 and 12.70 mm in the coniferous and broadleaved forests, respectively. Moreover, 44.3% and 46.7% of throughfall events had a total event precipitation amounts of less than 5 mm in the coniferous and broadleaved forests, respectively. There was no difference in the throughfall characteristics ($p > 0.1$) between the forest types.

Soil water potential at 10 cm showed the largest variation compared to other depths during the monitoring

TABLE 1 Information for soil water potential sensors installation depth and the corresponding soil properties (mean ± standard deviation) in two forest types

Slope position	Elevation (m)	Slope (°)	Depth (cm)	Bulk density (g/cm ³)	Porosity (%)	Soil organic matter (g/kg)	Clay content (%)	Rock fragments (%)	Root biomass (g/m ³)	Saturated hydraulic conductivity (cm/min)
Coniferous forest (<i>A. fabri</i> , <i>P. brachytyla</i>)										
Up slope	2287	25	10-50	0.86 ± 0.16*	0.67 ± 0.06*	48.32 ± 16.64	1.74 ± 0.02	9.66 ± 2.65 [§] *	113	0.89
			50-100	1.20 ± 0.11	0.55 ± 0.04 [§]	32.35 ± 0.20	1.37 ± 0.35	15.50 ± 2.50	NaN	0.10
Mid slope	2270	20	10-50	0.93 ± 0.20	0.65 ± 0.07	52.29 ± 11.58	1.22 ± 0.49	6.63 ± 2.60 [§] *	96	0.77
			50-100	1.05 ± 0.06 [§]	0.61 ± 0.02 [§]	28.75 ± 3.01	1.28 ± 0.32	10.44 ± 0.83	12	0.82
Down slope	2250	15	10-50	0.88 ± 0.13*	0.67 ± 0.05*	57.82 ± 27.01	2.29 ± 0.52	7.10 ± 0.66 [§] *	122	1.84
			50-100	1.06 ± 0.09 [§]	0.60 ± 0.04 [§]	33.26 ± 0.37	1.50 ± 0.76	10.17 ± 3.17	20	0.36
Broadleaved forest (<i>L. cleistocarpus</i> , <i>S. babylonica</i>)										
Up slope	3070	30	10-50	0.92 ± 0.20*	0.65 ± 0.08*	46.55 ± 16.42	3.15 ± 0.57	8.90 ± 2.16 [§] *	232	0.89
			50-100	1.44 ± 0.19	0.46 ± 0.07	31.54 ± 0.16	1.23 ± 0.65	11.66 ± 3.54	NaN	NaN
Mid slope	3045	24	10-50	0.89 ± 0.31*	0.66 ± 0.12*	50.43 ± 14.31	2.36 ± 0.71	5.10 ± 3.30 [§] *	202	0.89
			50-100	1.61 ± 0.10	0.39 ± 0.04	34.57 ± 1.65	1.26 ± 0.45	14.75 ± 3.66	13	0.18
Down slope	3025	18	10-50	0.96 ± 0.29*	0.64 ± 0.11*	59.59 ± 0.66	3.55 ± 1.48	5.62 ± 1.08 [§] *	212	0.34
			50-100	1.63 ± 0.09	0.39 ± 0.03	46.59 ± 7.21	1.66 ± 0.88	12.92 ± 4.53	22	0.85

*Denotes significant difference between soil layers.

[†]Denotes significant difference between forest types.

NaN denotes data not available.

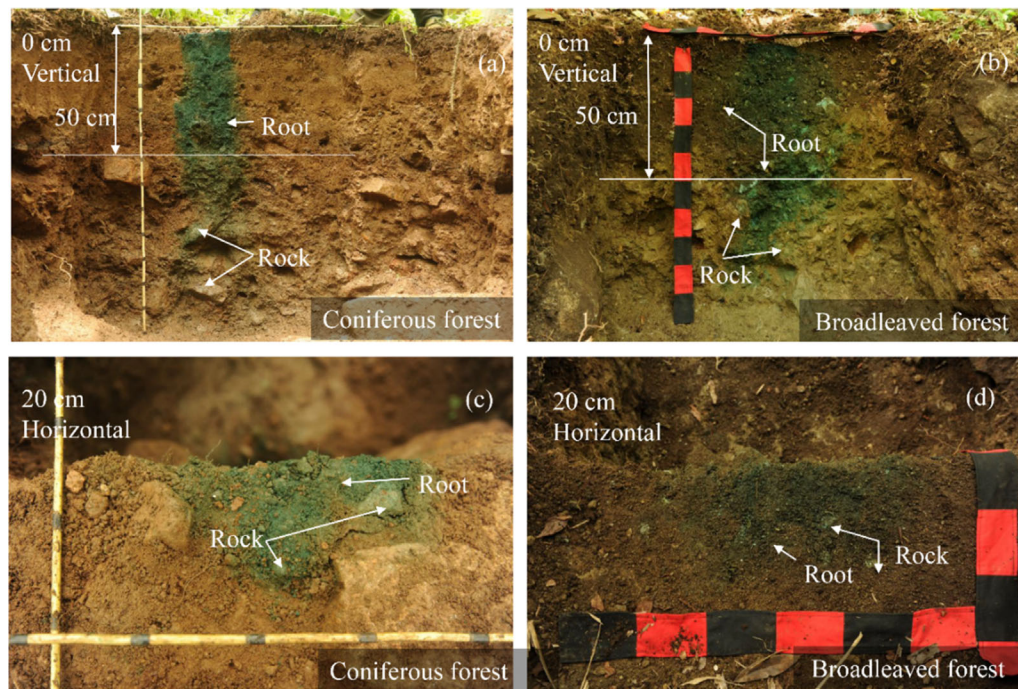


FIGURE 6 Photographs of dye-stained areas in the coniferous (a, c) and broadleaved forests (b, d) in horizontal and vertical directions

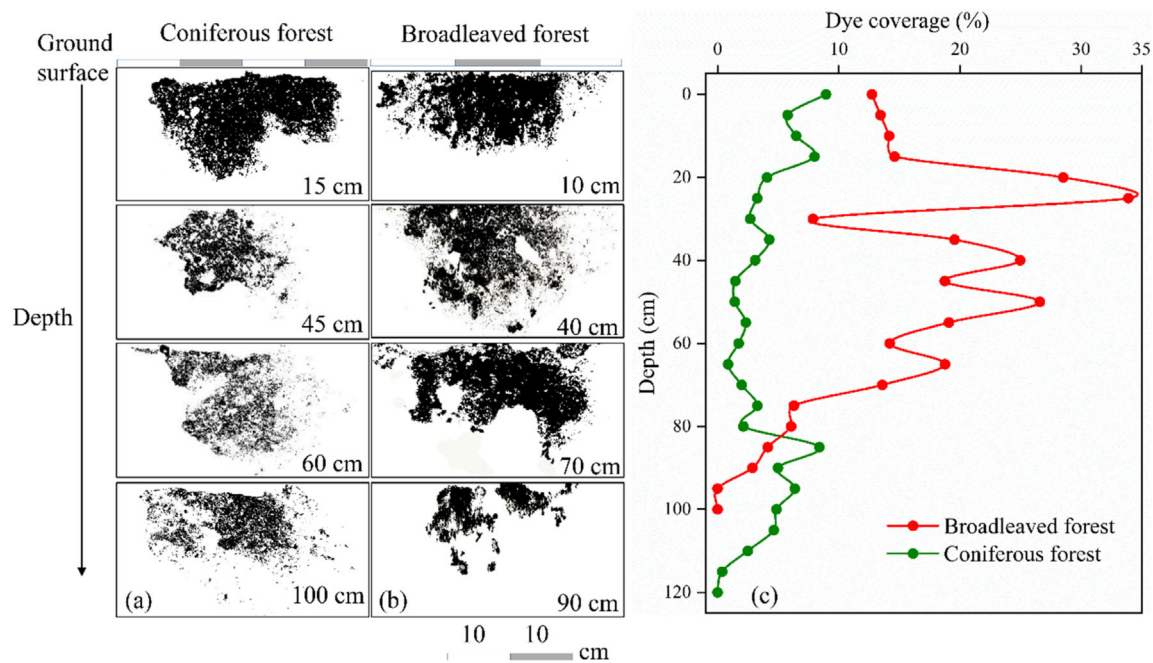


FIGURE 7 Binary digital images of dye-stained areas in the coniferous (a) and broadleaved forests (b), and dye coverage variation along the soil depth (c)

period in both forests. Soil water potential at 100 cm in the broadleaved forest experienced more intense variation compared with that in the coniferous forest, this indicated that intense interactions in the broadleaved forest may be the result of frequent PF occurrence. It should be noted that the soil water potential in the growing season was in high water potential status regardless of forest types and depths (Figure 9).

3.4 | Preferential flow frequency and controlling factors

There was a total of 128 (including 17 sequential PF events) and 91 (including 11 sequential PF events) events identified as PF, accounting for 67.8% and 71.7% of total response events in the coniferous and broadleaved forests, respectively, indicating that PF occurrence was

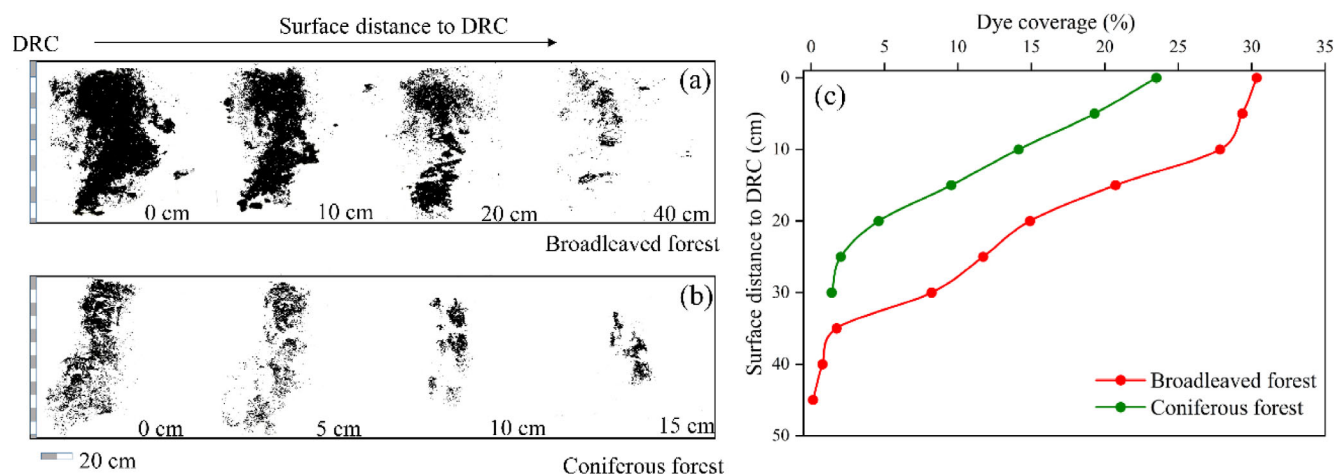


FIGURE 8 Binary digital images of dye-stained areas in the coniferous (a) and broadleaved forests (b), and dye coverage variation in the horizontal direction (c)

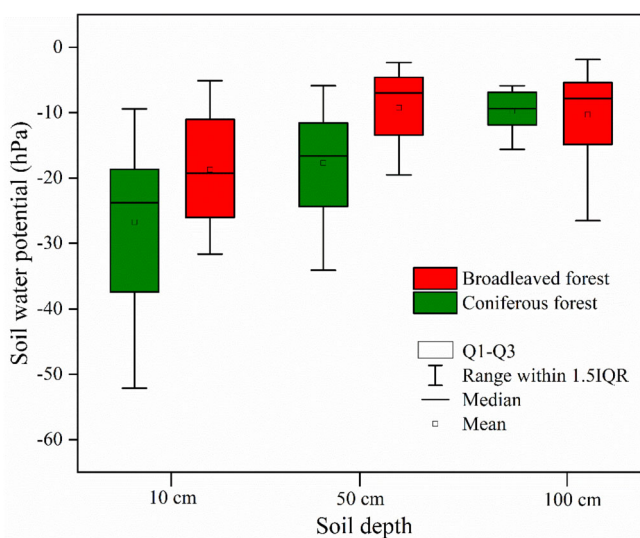


FIGURE 9 Mean soil water potential (mean value of all slope positions) in the coniferous and broadleaved forests during the growing season

common in our study. The minimum PF frequency was found at the upslope locations with 60.0% in the coniferous forest and 65.9% in the broadleaved forest, and the maximum PF frequency was found at the downslope locations in both forests with 74.6% in the coniferous forest and 79.1% in the broadleaved forest. Both forests showed an increasing trend of PF frequency from the upslope to downslope locations (Figure 10). Total throughfall amount and maximum throughfall intensity were identified as controlling factors for PF occurrence at the downslope locations in both forests (Table 2). A higher throughfall amount (21.87 mm for PF vs. 8.60 mm for SR in the coniferous forest; 15.65 mm for PF vs. 3.73 mm for SR in the broadleaved forest) and

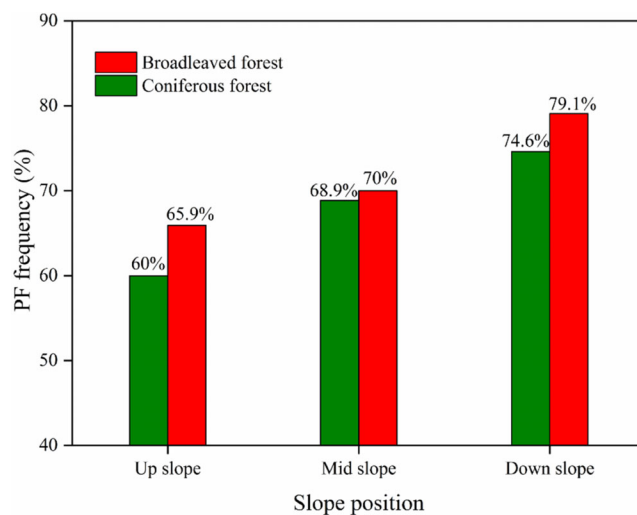


FIGURE 10 The PF frequency at different slope positions in the coniferous and broadleaved forests

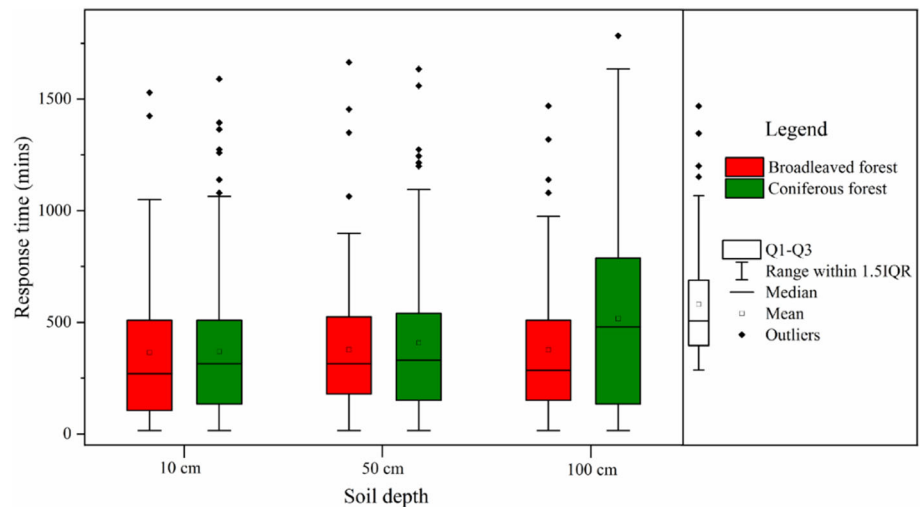
maximum throughfall intensity (4.63 mm/h for PF vs. 2.54 mm/h for SR in the coniferous forest; 4.46 mm/h for PF vs. 2.38 mm/h for SR in the broadleaved forest) was conducive to PF occurrence. Throughfall duration in the PF-producing events at the downslope location in the coniferous forest was much longer than that of SR events. Initial soil water conditions were not recognised as controlling factors in both forests. The response times of PF and SR events to throughfall events at different soil depths and forest types were compared (Figure 11). There was no significant difference ($p > 0.05$) in response times at 10 and 50 cm between the coniferous and broadleaved forests. A significant difference ($p < 0.05$) in response time at 100 cm was observed with the mean value of 517 mins in the coniferous forest and 365 mins in the broadleaved forest.

TABLE 2 Controlling factors of throughfall characteristics and initial soil moisture conditions that regulate preferential flow (PF) and sequential response (SR) events. Factors with significant p values (95% confidence interval) from t -tests are bold

Factors	Coniferous forest				Broadleaved forest			
	U	M	D	EH	U	M	D	EH
Throughfall amount (mm)	0.50	0.77	0.05	0.40	0.25	0.93	0.02	0.35
Duration (hour)	0.46	0.41	0.03	0.30	0.36	0.84	0.06	0.53
Mean intensity (mm/hour)	0.16	0.99	0.14	0.44	0.31	0.71	0.10	0.54
Maximum intensity (mm/hour)	0.22	0.59	0.04	0.58	0.94	0.53	0.01	0.82
Skewness	0.59	0.99	0.52	0.74	0.10	0.60	0.14	0.53
Kurtosis	0.31	0.93	0.43	0.55	0.15	0.53	0.14	0.54
Standard deviation	0.22	0.73	0.21	0.68	0.57	0.56	0.16	0.73
Initial SWP 10 cm (hPa)	0.53	0.12	0.67	0.19	0.54	0.85	0.77	0.28
Initial SWP 50 cm (hPa)	0.49	0.79	0.66	0.41	0.27	0.39	0.41	0.71
Initial SWP 100 cm (hPa)	0.27	0.74	0.54	0.86	0.73	0.12	0.11	0.93
Average profile SWP (hPa)	0.90	0.30	0.60	0.25	0.50	0.73	0.69	0.91

Abbreviations: SWP denotes soil water potential; U, M, D, and EH denote upslope, midslope, downslope, and entire hillslope, respectively.

FIGURE 11 Box plot of the response times of soil water potential to throughfall events in different depths and forest types



Mean PF frequency was 38.1% in the upper layer and 34.3% in the deeper layer in the coniferous forest, and 34.5% in the upper layer and 48.2% in the deeper layer in the broadleaved forest, and this demonstrated that the main discrepancy in PF frequency between forest types was attributed to the deeper soil layer. There was an increasing trend of PF frequency of the deeper soil layer from upslope to downslope locations in both forests (Figure 12). Factors of temporally varying factors (throughfall characteristics and initial soil water conditions) were not identified as controlling factors on PF occurrence between soil layers except the initial SWP of 50 cm at the hillslope scale (Table 3). The matrix conditions between the upper and deeper soil layers showed significant differences (see 2.1), indicating the dominant contribution of soil properties to PF occurrence between soil layers.

4 | DISCUSSION

4.1 | Spatial controlling factors on PF occurrence

The average PF frequency in the upper layer (10–50 cm) was 38.1% in the coniferous forest. This is comparable to the results from the Wüstebach catchment dominated by Norway spruce (*Picea abies* [L.]) and Sitka spruce (*Picea sitchensis*), where PF frequency ranged from 7% to 51%, and the maximum depth of sensor installation is 40 cm (Wiekenkamp et al., 2016). The average PF frequency in the broadleaved forest was 34.53%, which is similar to results observed in the deciduous Shale Hills forest (ranging from 16% to 47%), where the deepest sensors are installed at a depth of 50 cm (Liu & Lin, 2015). In the upper soil layer (10–50 cm), the root networks (Figure S1)

and hydrophobicity caused by the SOM usually led to high PF occurrence and subsurface flow in forest soils (Alaoui et al., 2011; Bogner et al., 2010; Gerke et al., 2015). A dense root network supplies abundant soil-root interfaces and decayed root channels to favour PF occurrence (Gerke et al., 2015). Higher SOM in the upper soil layer results in a stronger hydrophobicity effect, and the formation of organic coatings on soil aggregates makes soil hard to wet and forces water to converge into PF pathways (Gerke et al., 2015; Nimmo, 2012). The low clay content (<4%) was below the threshold of 8% for initiating and persisting macropore structures in soils (Koestel et al., 2012), indicating that the root networks constituted the main flow pathways in the

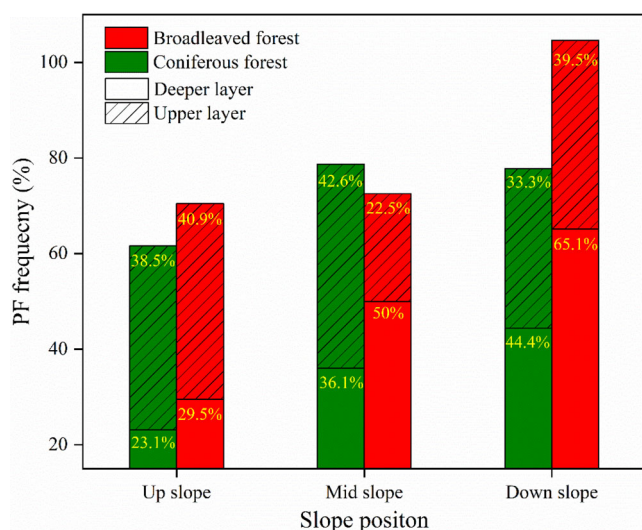


FIGURE 12 The PF frequency of the upper and deeper soil layers at different slope positions in the coniferous and broadleaved forests

upper soil layer in both forests (Bogner et al., 2010; Demand et al., 2019). Abrupt change of soil K_s in a short distance was observed in the upper soil layer, which is also favours PF in soils (Kishel & Gerla, 2002). Further, small-sized rock fragments in the upper layer might not strongly affect water flow since these rocks and surrounding soils were completely stained (Hartmann et al., 2020). Significant differences in root biomass between forest types resulted in a slight discrepancy in PF frequency in the upper soil layer, and this indicated that PF occurrence is probably related to fine roots (Luo et al., 2019).

Within the deeper soil layer (50–100 cm), mean PF frequency was 35.4% in the coniferous forest, and 48.2% in the broadleaved forest. Rock fragment contents in the deeper layer showed no difference between forest types, whereas less soil porosity in the broadleaved forest had higher PF occurrence, elucidating the fact that PF in stony soils uses only a small fraction of porosity (Clothier et al., 2008; Jarvis et al., 2016). The appearance of large rocks excluded in this study due to limited sampling scale offered fast PF pathways and retained limited rainfall water owing to the impermeability of rocks. In the broadleaved forest, rocks experienced more severe weathering than the coniferous forest owing to better thermal conditions resulting from its lower elevation, and more soil-rock interphases contributed to the generation of the funnel flow (Cerdà, 2001; Hartmann et al., 2020; Sohrt et al., 2014). Overall, rock fragments contributed to the development of continuous macropores and cracks for water flow, and are not a negligible component of PF occurrence (Li et al., 2019; Zhang et al., 2016).

Topographic effects on PF occurrence were evident in this study. An increasing trend of PF frequency from the upslope to downslope locations was observed on both

Factors	Coniferous forest				Broadleaved forest			
	U	M	L	EH	U	M	L	EH
Throughfall amount(mm)	0.58	0.95	0.23	0.60	0.58	0.95	0.23	0.23
Duration (hour)	0.73	0.89	0.32	0.44	0.73	0.89	0.32	0.22
Mean intensity (mm/hour)	0.96	0.82	0.60	0.23	0.96	0.82	0.60	0.50
Maximum intensity (mm/hour)	0.68	0.64	0.55	0.69	0.68	0.64	0.55	0.93
Skewness	0.77	0.14	0.80	0.59	0.77	0.14	0.80	0.50
Kurtosis	0.91	0.15	0.74	0.62	0.91	0.15	0.74	0.81
Standard deviation	0.92	0.66	0.62	0.43	0.92	0.66	0.62	0.71
Initial SWP 10 cm(hPa)	0.24	0.98	0.56	0.26	0.24	0.98	0.56	0.14
Initial SWP 50 cm (hPa)	0.38	0.93	0.50	0.28	0.38	0.93	0.50	0.04
Initial SWP 100 cm (hPa)	0.55	0.10	0.66	0.69	0.55	0.10	0.66	0.38
Average profile SWP(hPa)	0.24	0.99	0.56	0.93	0.24	0.99	0.56	0.08

Abbreviations: SWP denotes soil water potential; U, M, D, and EH denote upslope, midslope, downslope, and entire hillslope, respectively.

TABLE 3 Controlling factor of preferential flow (PF) occurrence in the upper (10–50 cm) and deeper soil layers (50–100 cm), respectively. Factors with significant p values (95% confidence interval) from t -tests are bold

forested hillslopes, similar to observations in other studies (Guo & Lin, 2018; Holden, 2009; Öhrström et al., 2002). The higher hydrophobicity caused by higher SOM, gentler slope and increased contributing area at downslope locations enhanced PF occurrence in both forests (Benegas et al., 2014; Guo & Lin, 2018; Hardie et al., 2011). Topography seemed not to have an effect on the PF frequency in the upper soil layer (Figure 12), which is characterised by high heterogeneity caused by complex root networks in both forests.

4.2 | Temporal controlling factors on PF occurrence

Throughfall characteristics, especially its intensity, were believed to be the key factors controlling PF occurrence (Demand et al., 2019; Green & Erskine, 2011; Hopkins et al., 2016; Liu & Lin, 2015; Tian et al., 2019), and a positive relationship between throughfall intensity and PF frequency was also found in our study (Figure S2). Total throughfall amount and maximum throughfall intensity were identified as controlling factors for PF occurrence ($p < 0.05$) at the downslope locations in both forests (Graham & Lin, 2011; Liu & Lin, 2015). This was the combined effect of topography and throughfall. Throughfall characterised by long duration and low intensity tends to moisten soil more uniformly, creating prerequisite for subsurface connectivity. Besides, gentler slopes favour the formation of the transient water table at layers with contrasting hydraulic conductivity, and forces water flow into PF pathways at the downslope locations (Guo & Lin, 2018).

Initial soil water conditions were not identified as controlling factors in both forests, and this is consistent with Nimmo (2012) who argued that local saturation of soil was more likely to drive PF regardless of initial soil water conditions. Similarly, Wiekenkamp et al. (2016) found no clear relationship between the PF frequency and initial soil water conditions when rainfall amount was less than 25 mm, and Tang et al. (2020) found a significant effect of initial soil water conditions on PF occurrence at only one of three studied sites. Notably, soil moisture showed slight fluctuation owing to plentiful and concentrated throughfall during the monitoring period and was possibly far from the threshold of influencing PF occurrence (Figure S3).

4.3 | Integrating multiple methods for PF investigation

Soil moisture monitoring network has created new opportunities for detecting and quantifying PF with high

spatial and temporal resolution. This method has the advantage of being noninvasive or minimally invasive and is a real-time reflection of soil moisture to dynamic throughfall input with the ability to bridge the gaps between scales and predict PF patterns in soils (Guo & Lin, 2018; Jarvis et al., 2016). Meanwhile, validation of PF information revealed by the soil moisture monitoring network and possible PF pathways are essential for a better understanding of PF.

Dye tracing experiments can supply two-dimensional images of spatial variations in water flow patterns at the pedon scale (Hartmann et al., 2020; Jarvis, 2007; Weiler & Flühler, 2004). Visualisation of flow pathways is based on image information interpreted by photographing excavated soil profiles, and detailed soil profile information (e.g., soil structure, distributions of roots and rocks) can be observed and obtained at the same time to connect PF pathways with soil properties or soil structure (Gerke et al., 2015; Guo & Lin, 2018; Holden, 2009). Nevertheless, the dye tracing experiment obtained limited information about the real-time processes and their temporal dynamics, which can be compensated for with information gained by a soil moisture monitoring network. Thus, the combination of the two methods enables exploring the theoretical mechanisms of PF (Guo & Lin, 2018; Jarvis et al., 2016). For example, soil moisture monitoring revealed higher PF frequency in the deeper soil layer at the downslope location in the coniferous forest, while soil properties showed no dramatic difference among the slope positions. In this situation, the dye tracing experiment at the downslope location revealed the mechanism behind it, demonstrating that the high PF occurrence was attributed to large rocks, which dispersed water flow through lacunar as PF pathways to the deeper soil layer. The agreement between results derived from the soil moisture monitoring network and dye tracing experiments confirmed the important effect of rock fragments on PF occurrence. Without the dye tracing experiment, the soil moisture monitoring network cannot reveal the possible flow pathways and dye tracing experiments cannot link the lacunar (soil-rock interfaces) with PF frequency. Therefore, combining soil moisture monitoring networks and dye tracing experiments enhances a more reliable interpretation.

5 | CONCLUSIONS

The hydrological process of PF is challenging to study, given the need to characterise vast heterogeneities in hydraulic properties of subsurface flow environments and changing environmental conditions over time. Our study integrated a soil moisture monitoring network and

dye tracing experiments, providing an important view of PF pathways and frequency. Results showed that the root network and rock fragments are of importance in PF occurrence on forested hillslopes of the eastern Tibetan Plateau. Specifically, the root networks, rock fragment content, and coarse-textured soils confirmed the high frequency of PF occurrence in the coniferous and broad-leaved forests along the elevation of the Gongga Mountain. Focus on the comparison of the soil layers and slope positions was addressed, demonstrating that the frequency of PF in the deeper soil layer (i.e., stony saprolite soil) increased from upslope to downslope locations on both forested hillslopes, highlighting the tendency for PF to occur downslope. In contrast, PF frequency in the upper soil layer showed slight change between forest types due to high heterogeneity caused by abundant root networks.

The integration of dye tracing experiments (which provides information on PF pathways) and a high-frequency soil moisture monitoring network (which provides temporal dynamics and spatial variation of PF) is a promising way forward in the investigation of PF and its controlling factors under field conditions. Future directions towards the operational prediction of PF in forested landscapes might focus on the assimilation, synthesis, and integration of high-resolution data on climatic forcings (e.g., of precipitation and throughfall) and subsurface architecture and properties (e.g., of the distribution of pore spaces and characteristics of the porous media). This information, in turn, could be used in concert with a high-frequency soil moisture monitoring network to capture the dynamic field conditions (Guo & Lin, 2018). The time is ripe for such high-resolution studies at hillslope and watershed scales, given advances in characterising surficial topography (Brubaker et al., 2013), scaling micro-computed tomography (CT) analyses to field settings, and in geophysical characterisation of deep critical zone structure using novel approaches (Flinchum et al., 2018; Oakley et al., 2021).

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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